TARGET OF A FINITE SURJECTION FROM AN AFFINE SPACE IS AFFINE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 3598-3638, following the statement at 3591-3596.

Editorial note. No load-bearing omissions; the final step 'Y_i affine cofinally implies Y affine' uses the affine-transition limit formalism.

Allowed standard prerequisites (editorial).

- Finite morphisms are affine and universally closed. Morphisms of Spaces, Lemma spaces-morphisms-lemma-integral-universally-closed (cross-chapter, cited at ~3602-3604). f is affine and universally closed.
- Image of universally closed from separated source. Morphisms of Spaces, Lemma spaces-morphisms-lemma-image-universally-closed-separated (cross-chapter, cited at ~3605-3607). Y is separated.
- Finite inside finite and finite presentation. lemma-finite-in-finite-and-finite-presentation (same chapter, cited at ~3612-3614). X = lim X_a with X_a -> Y finite and of finite presentation.
- Affineness of the limit descends. lemma-limit-is-affine (same chapter, card L26293, cited at ~3615-3616 and ~3632-3633). X_a (resp. X_i) affine for large index.
- Approximation by finite presentation spaces. proposition-approximate (same chapter, published as problem 1743 tag 07SU, cited at ~3621-3623). Y = lim Y_i with Y_i of finite presentation over Z.
- Descent of finite presentation morphisms. lemma-descend-finite-presentation (same chapter, card L26286, cited at ~3624-3627). X_0 -> Y_0 fp with X_i = X_0 x_{Y_0} Y_i and X = lim X_i.
- Descent of finiteness and surjectivity. lemma-descend-finite and lemma-descend-surjective (same chapter, cited at ~3628-3631). X_i -> Y_i finite and surjective for i large.
- Image of finite affine is affine. Cohomology of Spaces, Lemma spaces-cohomology-lemma-image-affine-finite-morphism-affine-Noetherian (cross-chapter, cited at ~3634-3635). Y_i affine.

Source reference. Original lemma 084Z: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 1865–1865. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 07SK: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 2148–2148. Navigation pointer only: the counted proof cites

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this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 07SN: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 1776–1776. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 07VR: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 2961–2961. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 07SQ: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 1370–1370. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original proposition 07SU: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 2328–2328. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 07VP: focused-seeds/geometry/S9-spaces-limits-affine/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03I5: focused-seeds/geometry/S9-spaces-limits-affine/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 05Z2: focused-seeds/geometry/S9-spaces-limits-affine/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0415: focused-seeds/geometry/S9-spaces-limits-affine/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Assume that f is surjective and finite, and assume that X is affine. Then Y is affine.*

Proof. We may and do view $f : X \to Y$ as a morphism of algebraic spaces over $\mathrm{Spec}(\mathbf{Z})$ (see Spaces, Definition 03I5). Note that a finite morphism is affine and universally closed, see Morphisms of Spaces, Lemma 0415. By Morphisms of Spaces, Lemma 05Z2 we see that Y is a separated algebraic space. As f is surjective and X is quasi-compact we see that Y is quasi-compact.

By Lemma 07VR we can write $X = \lim X_a$ with each $X_a \to Y$ finite and of finite presentation. By Lemma 07SQ we see that X_a is affine for a large enough. Hence we may and do assume that $f : X \to Y$ is finite, surjective, and of finite presentation.

By Proposition 07SU we may write $Y = \lim Y_i$ as a directed limit of algebraic spaces of finite presentation over $\mathbf{Z}$. By Lemma 07SK we can find $0 \in I$ and a morphism $X_0 \to Y_0$ of finite presentation such that $X_i = X_0 \times_{Y_0} Y_i$ for $i \geq 0$ and such that $X = \lim_i X_i$. By Lemma 084Z we see that $X_i \to Y_i$ is finite for i large enough. By Lemma 07SN we see that $X_i \to Y_i$ is surjective for i large enough. By Lemma 07SQ we see that X_i is affine for i large enough. Hence for i large enough we can apply Cohomology of Spaces, Lemma 07VP to conclude that Y_i is affine. This implies that Y is affine and we conclude. □